

# SET & JOINS

## QUESTION 1 - CROSS JOIN

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT *  
FROM first_table CROSS JOIN second_table
```

The Results tab displays the output of the query, showing a list of rows with columns: id, name, address. The first three rows are:

| id | name | address |
|----|------|---------|
| 1  | John | 1234    |
| 2  | John | 1234    |
| 3  | John | 1234    |

The Action Output tab shows the execution plan for the query, indicating that the CROSS JOIN operation is performed between first\_table and second\_table.

## QUESTION 2 - INNER JOIN

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT *  
FROM first_table  
INNER JOIN second_table  
ON first_table.id = second_table.id
```

The Results tab displays the output of the query, showing a list of rows with columns: id, name, address. The first three rows are:

| id | name | address |
|----|------|---------|
| 1  | John | 1234    |
| 2  | John | 1234    |
| 3  | John | 1234    |

The Action Output tab shows the execution plan for the query, indicating that the INNER JOIN operation is performed between first\_table and second\_table.

## QUESTION 3 - INNER JOIN

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT *  
FROM first_table  
INNER JOIN second_table  
ON first_table.id = second_table.id
```

The Results tab displays the output of the query, showing a list of rows with columns: id, name, address. The first three rows are:

| id | name | address |
|----|------|---------|
| 1  | John | 1234    |
| 2  | John | 1234    |
| 3  | John | 1234    |

The Action Output tab shows the execution plan for the query, indicating that the INNER JOIN operation is performed between first\_table and second\_table.

## QUESTION 4 - NATURAL JOIN

MySQL Workbench

2520090181\_himanees

SQL Editor

4 sessions

Practical-7

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

| id | name | address |
|----|------|---------|
| 1  | alex | DELHI   |
| 2  | adam | MUMBAI  |
| 3  | alex | CHENNAI |

Result 25

Action Output

| #  | Time     | Action                    | Message           | Duration / Feat       |
|----|----------|---------------------------|-------------------|-----------------------|
| 43 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 44 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 45 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 46 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 47 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 48 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 49 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 50 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 51 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 52 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 53 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 54 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 55 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 5 - LEFT JOIN

MySQL Workbench

2520090181\_himanees

SQL Editor

4 sessions

Practical-7

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

| id | name | id | address |
|----|------|----|---------|
| 1  | alex | 1  | DELHI   |
| 2  | adam | 2  | MUMBAI  |
| 3  | alex | 3  | CHENNAI |
| 4  | alex | 4  | CHENNAI |
| 5  | alex | 5  | CHENNAI |

Result 26

Action Output

| #  | Time     | Action                    | Message           | Duration / Feat       |
|----|----------|---------------------------|-------------------|-----------------------|
| 44 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 45 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 46 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 47 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 48 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 49 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 50 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 51 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 52 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 53 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 54 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 55 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 6 - LEFT JOIN

MySQL Workbench

2520090181\_himanees

SQL Editor

4 sessions

Practical-7

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

| id | name | id | address |
|----|------|----|---------|
| 4  | alex | 1  | DELHI   |
| 5  | alex | 2  | MUMBAI  |
| 6  | alex | 3  | CHENNAI |
| 7  | alex | 4  | CHENNAI |
| 8  | alex | 5  | CHENNAI |

Result 27

Action Output

| #  | Time     | Action                    | Message           | Duration / Feat       |
|----|----------|---------------------------|-------------------|-----------------------|
| 45 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 46 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 47 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 48 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 49 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 50 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 51 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 52 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 53 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 54 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 55 | 13.40.54 | SELECT * FROM first_table | 1 row(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 7 - RIGHT JOIN

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class_id = class_info_id
```

The Results grid displays the following data:

| id | name  | id | address |
|----|-------|----|---------|
| 1  | dev   | 1  | DELHI   |
| 2  | adnan | 2  | MUMBAI  |
| 3  | dev   | 3  | CHENNAI |
| 4  | dev   | 7  | NOWA    |
| 5  | dev   | 8  | PANIPAT |

The Action Output pane shows the execution plan for the query, including the following steps:

- 46 13.40.54 SELECT \* FROM class
- 47 13.40.54 SELECT \* FROM class\_info
- 48 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 49 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 50 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 51 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 52 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
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- 54 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 55 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 56 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 57 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id

## QUESTION 8 - RIGHT JOIN

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class_id = class_info_id
WHERE class_id IS NULL
```

The Results grid displays the following data:

| id | name    | id | address |
|----|---------|----|---------|
| 7  | NOWA    |    |         |
| 8  | PANIPAT |    |         |

The Action Output pane shows the execution plan for the query, including the following steps:

- 47 13.40.54 SELECT \* FROM class
- 48 13.40.54 SELECT \* FROM class\_info
- 49 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 50 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 51 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 52 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
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- 54 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 55 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 56 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 57 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id

## QUESTION 9 - FULL OUTER JOIN

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT *
FROM class
FULL OUTER JOIN class_info
ON class_id = class_info_id
```

The Results grid displays the following data:

| id | name  | id | address |
|----|-------|----|---------|
| 1  | dev   | 1  | DELHI   |
| 2  | adnan | 2  | MUMBAI  |
| 3  | dev   | 3  | CHENNAI |
| 4  | dev   | 7  | NOWA    |
| 5  | dev   | 8  | PANIPAT |

The Action Output pane shows the execution plan for the query, including the following steps:

- 49 13.40.54 SELECT \* FROM class
- 50 13.40.54 SELECT \* FROM class\_info
- 51 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 52 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 53 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 54 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 55 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 56 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 57 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 58 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id
- 59 13.40.54 SELECT \* FROM class\_info WHERE class\_id = class\_info\_id

## QUESTION 10 - FULL OUTER JOIN

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class_id = class_info.id
UNION
SELECT *
FROM class
```

The results grid displays the following data:

| id | name | id | address |
|----|------|----|---------|
| 1  | adam | 1  | DELHI   |
| 2  | adan | 2  | MUMBAI  |
| 3  | alex | 3  | CHENNAI |
| 4  | alex | 7  | KOCHI   |
| 5  | alex | 8  | PANPAT  |

The Action Output pane shows the execution details for the query, including the time taken and the number of rows returned.

## QUESTION 11 – UNION

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT *
FROM first_table
UNION
SELECT *
FROM second_table
```

The results grid displays the following data:

| id | name    |
|----|---------|
| 1  | adan    |
| 2  | adan    |
| 3  | chester |

The Action Output pane shows the execution details for the query, including the time taken and the number of rows returned.

## QUESTION 12 – UNION

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
SELECT name
FROM first_table
UNION
SELECT name
FROM second_table
```

The results grid displays the following data:

| name    |
|---------|
| adan    |
| adan    |
| chester |

The Action Output pane shows the execution details for the query, including the time taken and the number of rows returned.

## QUESTION 13 - UNION AL

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT *  
FROM first_table  
UNION ALL  
SELECT *  
FROM second_table
```

The Results grid shows the output of the query, which consists of two tables. The first table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea'). The second table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea').

The Action Output pane shows the execution log, including the query text and the number of rows returned for each statement.

## QUESTION 14 - UNION AL

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
UNION ALL  
SELECT *  
FROM second_table  
SELECT *  
FROM first_table
```

The Results grid shows the output of the query, which consists of two tables. The first table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea'). The second table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea').

The Action Output pane shows the execution log, including the query text and the number of rows returned for each statement.

## QUESTION 15 - INTERSECT

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT *  
FROM first_table  
INTERSECT  
SELECT *  
FROM second_table
```

The Results grid shows the output of the query, which consists of two tables. The first table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea'). The second table has columns 'id' and 'name' with values (1, 'adam'), (2, 'adam'), and (3, 'chelsea').

The Action Output pane shows the execution log, including the query text and the number of rows returned for each statement.

## QUESTION 16 – INTERSECT

The screenshot shows MySQL Workbench with a query editor containing the following SQL code:

```
SELECT f.name  
FROM first_table f  
WHERE f.name = s.name  
INTERSECT  
SELECT f.*
```

The query is executed, and the output shows a single row with the name 'adam'.

| name |
|------|
| adam |

The bottom pane shows the execution log with the following entry:

| Time     | Action                                                                              | Message            | Duration / Fetch      |
|----------|-------------------------------------------------------------------------------------|--------------------|-----------------------|
| 13.47:03 | SELECT * FROM class RIGHT JOIN class_info ON class_id = class_info.cl LIMIT 0, 1000 | 5 rows(s) returned | 0.000 sec / 0.000 sec |

## QUESTION 17 – MINUS

The screenshot shows MySQL Workbench with a query editor containing the following SQL code:

```
SELECT f.name  
FROM first_table f  
LEFT JOIN second_table s  
ON f.name = s.name  
WHERE s.name IS NULL  
MINUS  
SELECT c.id
```

The query is executed, and the output shows a single row with the name 'adam'.

| name |
|------|
| adam |

The bottom pane shows the execution log with the following entry:

| Time     | Action                                                                                                     | Message            | Duration / Fetch      |
|----------|------------------------------------------------------------------------------------------------------------|--------------------|-----------------------|
| 13.47:03 | SELECT * FROM class RIGHT JOIN class_info ON class_id = class_info.cl WHERE class_id IS NULL LIMIT 0, 1000 | 2 rows(s) returned | 0.015 sec / 0.000 sec |

## QUESTION 18 – MINUS

The screenshot shows MySQL Workbench with a query editor containing the following SQL code:

```
SELECT c.id,  
c.name,  
ci.address  
FROM class c  
INNER JOIN class_info ci  
ON c.id = ci.cl
```

The query is executed, and the output shows three rows with the following data:

| id | name | address |
|----|------|---------|
| 1  | adam | DELHI   |
| 2  | adam | MUMBAI  |
| 3  | alex | CHEMNAI |

The bottom pane shows the execution log with the following entry:

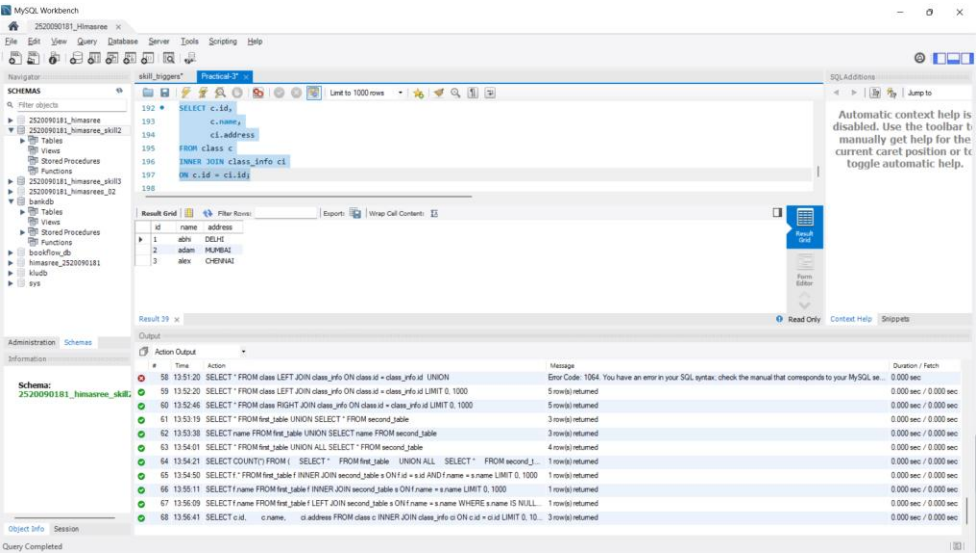
| Time     | Action                                                                     | Message                                                                                                         | Duration / Fetch |
|----------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|
| 13.51:20 | SELECT * FROM class LEFT JOIN class_info ON class_id = class_info.cl UNION | Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL version | 0.000 sec        |

QUESTION 19

SCENARIO:

Find students having matching addresses.

QUESTION 19



QUESTION 20

SCENARIO:

Generate a report showing all students and address availability status.

